

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458588.8795 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.029
Transit depth (Rp/Rs)^2	2.48 +/- 0.9 [%]
Orbital Inclination (inc)	88.76 +/- 2.51
Airmass coefficient 1 (a1)	1.037 +/- 0.022
Airmass coefficient 2 (a2)	-0.033 +/- 0.026
Scatter in the residuals of the lightcurve fit is	2.13 %
Best Comparison Star	#3 - [623, 248]
Optimal Aperture	4.3
Optimal Annulus	12.5
Transit Duration (day)	0.099 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.029 vs 0.126 (deviation 1.07σ)
Duration fit vs published	0.099 d vs 0.0925 d (deviation 1.81σ)
Mid-time O–C	2.3 min
Depth SNR	5.4
Residual scatter	2.13 %

Light curve

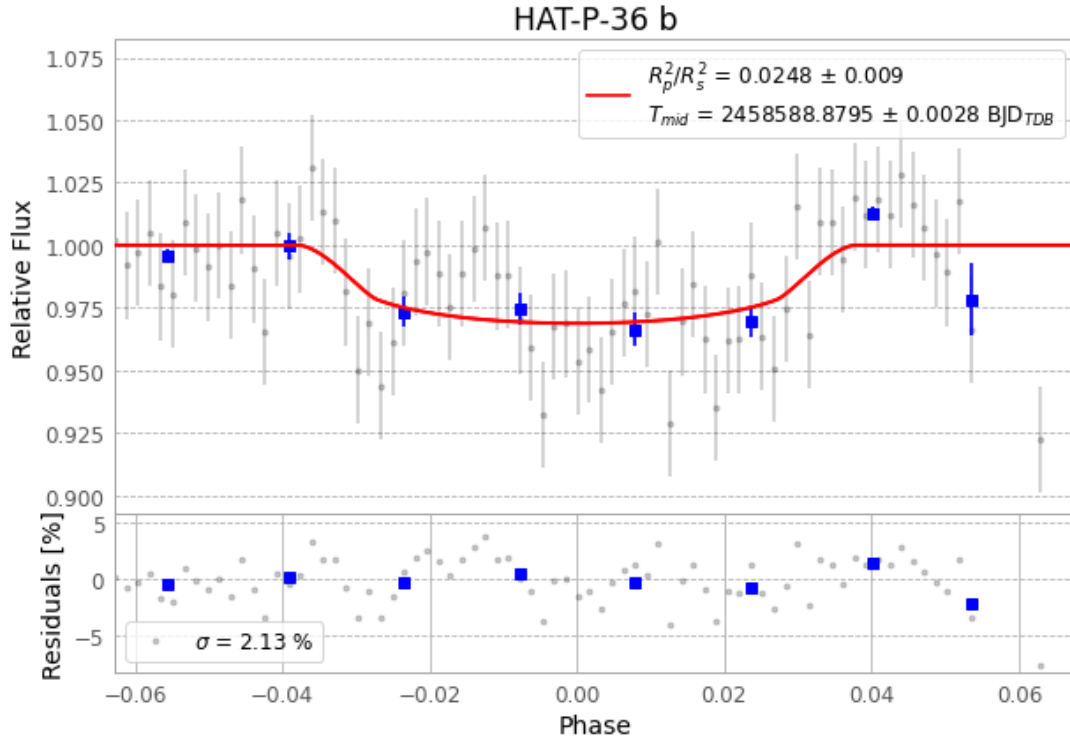


Figure 1 — FinalLightCurve_HAT-P-36 b_2019-04-15.png (SHA-256 6e0fff43ccb5891...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [398, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fe395fefd93f189c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), HATP-36190415070012.FITS → HATP-36190415111212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-190415.log (734ba56b78bd...), FinalLightCurve_HAT-P-36 b_2019-04-15.png (6e0fff43ccb...), FinalLightCurve_HAT-P-36 b_2019-04-15.csv (195ad114ae03...)

Compute resources

Wall time 225.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 20:33Z.